=== UNIT 2: SINGLE-PHASE AC CIRCUITS, PHASORS & RESONANCE ===

>> 2.1 Generation of Alternating Quantities and Sinusoidal Waveforms

Alternating Current (AC) is generated by rotating a conductor coil in a uniform magnetic field according to Faraday's Law of Electromagnetic Induction: e(t) = V_m * sin(omega * t + phi).
* Key Definitions: Instantaneous value (value at a specific time t), Peak value (maximum amplitude V_m), Cycle (one complete set of positive and negative values), Frequency f = 1/T (in Hertz), Angular Frequency omega = 2 * pi * f (in rad/sec).
* Average Value (V_avg): Mean value over half a cycle for symmetrical waveforms: V_avg = (2 / pi) * V_m ~ 0.637 * V_m.
* Root Mean Square (RMS / Effective) Value (V_rms): The equivalent DC value that produces identical heating in a resistor: V_rms = sqrt((1 / T) * int_0^T v(t)^2 dt) = V_m / sqrt(2) ~ 0.707 * V_m.
* Form Factor: Ratio of RMS value to Average value: Form Factor = V_rms / V_avg = (V_m / sqrt(2)) / (2*V_m / pi) = pi / (2*sqrt(2)) ~ 1.11.
* Peak (Crest) Factor: Ratio of Maximum value to RMS value: Peak Factor = V_m / V_rms = V_m / (V_m / sqrt(2)) = sqrt(2) ~ 1.414.

>> 2.2 Phasor Representation and Elementary AC Branches

* Pure Resistor: Voltage and current are strictly in phase (phase angle phi = 0). Resistance R is independent of frequency.
* Pure Inductor: Current lags voltage by exactly 90 degrees (pi/2 radians). Inductive Reactance X_L = omega * L = 2 * pi * f * L (in Ohms).
* Pure Capacitor: Current leads voltage by exactly 90 degrees (pi/2 radians). Capacitive Reactance X_C = 1 / (omega * C) = 1 / (2 * pi * f * C) (in Ohms).
* Series RLC Circuit: Total complex impedance Z = R + j * (X_L - X_C) = |Z| * angle(phi).
- Impedance Magnitude: |Z| = sqrt(R^2 + (X_L - X_C)^2).
- Phase Angle: phi = arctan((X_L - X_C) / R).
- Current: i(t) = I_m * sin(omega * t - phi), where I_m = V_m / |Z|.

>> 2.3 Series Resonance and Frequency Response

* Resonance Condition: Occurs when inductive reactance equals capacitive reactance: X_L = X_C.
omega_0 * L = 1 / (omega_0 * C) => omega_0^2 = 1 / (L * C) => f_0 = 1 / [2 * pi * sqrt(L * C)].
* Characteristics of Series Resonance:
1. Net reactance is zero; Impedance is at absolute minimum: Z_min = R (purely

resistive).
2. Current is at absolute maximum: I_0 = V / R, in phase with supply voltage (Power Factor = 1.0).
3. Quality Factor (Q): Magnification of reactive voltages across L and C relative to supply voltage:
Q = V_L / V = (I_0 * omega_0 * L) / (I_0 * R) = (omega_0 * L) / R = [1 / (omega_0 * C * R)] = (1 / R) * sqrt(L / C).
4. Bandwidth (BW): Range of frequencies where power >= P_max / 2 (half-power frequencies f1, f2):
BW = f2 - f1 = f_0 / Q = R / (2 * pi * L).

>> 2.4 AC Power Analysis: The Power Triangle

* Real (Active / True) Power (P): Power consumed in resistive heating: P = V_rms * I_rms * cos(phi) (in Watts [W] or kW).
* Reactive Power (Q): Power oscillating between magnetic/electric fields and source: Q = V_rms * I_rms * sin(phi) (in Volt-Amperes Reactive [VAR] or kVAR).
* Apparent Power (S): Total complex power delivered by source: S = V_rms * I_rms = sqrt(P^2 + Q^2) (in Volt-Amperes [VA] or kVA).
* Power Factor (PF): Cosine of phase angle: PF = cos(phi) = P / S = R / |Z|. Lagging for inductive loads, leading for capacitive loads.

[SOLVED PROBLEM 1: Series RLC Circuit Impedance and Power Factor]

Problem: A series circuit consists of a resistance R = 15 Omega, inductance L = 0.05 H, and capacitance C = 100 microF connected across a 230 V, 50 Hz AC supply. Calculate: (1) Inductive reactance X_L, (2) Capacitive reactance X_C, (3) Total impedance Z, (4) RMS current I, (5) Power Factor cos(phi), (6) Active Power P.
Solution Step 1: Compute Reactances at f = 50 Hz (omega = 2 * pi * 50 = 314.16 rad/s):
X_L = omega * L = 314.16 * 0.05 = 15.71 Omega.
X_C = 1 / (omega * C) = 1 / (314.16 * 100 * 10^-6) = 1 / 0.031416 = 31.83 Omega.
Solution Step 2: Total Net Reactance and Impedance:
Net Reactance X = X_L - X_C = 15.71 - 31.83 = -16.12 Omega (Capacitive).
|Z| = sqrt(R^2 + X^2) = sqrt(15^2 + (-16.12)^2) = sqrt(225 + 259.85) = sqrt(484.85) = 22.02 Omega.
Phase Angle: phi = arctan(-16.12 / 15) = arctan(-1.0747) = -47.06 degrees.
Solution Step 3: Compute Circuit Current:
I_rms = V_rms / |Z| = 230 / 22.02 = 10.45 Amperes.
Solution Step 4: Power Factor:
PF = cos(phi) = cos(-47.06 deg) = R / |Z| = 15 / 22.02 = 0.681 LEADING (since X_C > X_L).
Solution Step 5: Active Power Consumed:
P = V_rms * I_rms * cos(phi) = 230 * 10.45 * 0.681 = 1,637 Watts = 1.637 kW.

Check via I^2 * R: P = (10.45)^2 * 15 = 109.20 * 15 = 1,638 Watts (Confirmed).

[SOLVED PROBLEM 2: Series Resonance Frequency, Q-Factor & Voltage Magnification]

Problem: An RLC series circuit has R = 2 Omega, L = 1 mH (10^-3 H), and C = 0.1 microF (10^-7 F) excited by a 10V AC generator. Determine: (1) Resonant frequency f_0, (2) Quality factor Q, (3) Half-power bandwidth BW, (4) Voltage across inductor at resonance.
Solution Step 1: Calculate Resonant Frequency f_0:
f_0 = 1 / [2 * pi * sqrt(L * C)] = 1 / [2 * pi * sqrt(10^-3 * 10^-7)]
sqrt(10^-10) = 10^-5.
f_0 = 1 / [2 * pi * 10^-5] = 100,000 / (2 * pi) = 15,915.5 Hz ~ 15.92 kHz.
Solution Step 2: Calculate Quality Factor Q:
Q = (1 / R) * sqrt(L / C) = (1 / 2) * sqrt(10^-3 / 10^-7) = (1 / 2) * sqrt(10,000) = (1 / 2) * 100 = 50.
Solution Step 3: Calculate Bandwidth BW:
BW = f_0 / Q = 15,915.5 Hz / 50 = 318.3 Hz.
Solution Step 4: Current and Voltage Magnification at Resonance:
Current at resonance: I_0 = V / R = 10 V / 2 Omega = 5.0 Amperes.
Inductive reactance at resonance: X_L0 = 2 * pi * f_0 * L = 2 * pi * 15,915.5 * 10^-3 = 100 Omega.
Voltage across inductor: V_L = I_0 * X_L0 = 5.0 A * 100 Omega = 500 Volts.
Voltage Magnification Factor: V_L / V_supply = 500 V / 10 V = 50 = Q. (Q-factor magnifies terminal voltage 50-fold).

[SOLVED PROBLEM 3: Complete Power Triangle and Power Factor Correction]

Problem: An industrial biotechnology plant motor load draws 50 kW at 400 V, 50 Hz with a lagging power factor of 0.707. Determine: (1) Apparent power S1 and reactive power Q1, (2) Capacitance required in parallel to improve the overall power factor to 0.95 lagging.
Solution Step 1: Initial Electrical Operating State:
cos(phi1) = 0.707 => phi1 = 45.0 degrees => tan(phi1) = tan(45) = 1.0.
Active Power P = 50 kW.
Initial Reactive Power Q1 = P * tan(phi1) = 50 * 1.0 = 50 kVAR.
Initial Apparent Power S1 = P / cos(phi1) = 50 / 0.707 = 70.72 kVA.
Solution Step 2: Desired Operating State with Improved Power Factor:
cos(phi2) = 0.95 => phi2 = 18.19 degrees => tan(phi2) = tan(18.19) = 0.3287.
New Reactive Power Q2 = P * tan(phi2) = 50 * 0.3287 = 16.44 kVAR.
Solution Step 3: Required Reactive Power Provided by Shunt Capacitor:
Q_C = Q1 - Q2 = 50.0 - 16.44 = 33.56 kVAR = 33,560 VAR.
Solution Step 4: Compute Required Capacitance C:
Q_C = V^2 * omega * C => C = Q_C / (V^2 * 2 * pi * f)

C = 33,560 / [(400)^2 * 2 * pi * 50] = 33,560 / [160,000 * 314.16] = 33,560 / 50,265,482
C = 6.677 * 10^-4 Farads = 667.7 microF.
Conclusion: Installing a 667.7 microF shunt capacitor bank reduces kVA demand from 70.7 kVA to 52.6 kVA, lowering line current and eliminating utility low-PF penalties.

[SOLVED PROBLEM 4: AC Waveform Effective (RMS) and Form Factor Derivation]

Problem: Derive from first principles the RMS value, average value, and form factor of a symmetrical half-wave rectified sinusoidal voltage waveform v(t) = V_m * sin(omega * t) for 0 <= omega*t <= pi, and v(t) = 0 for pi <= omega*t <= 2*pi.
Solution Step 1: Calculate Average Value V_avg over full period T = 2*pi:
V_avg = (1 / 2*pi) * int_0^pi V_m * sin(theta) d(theta) = (V_m / 2*pi) * [-cos(theta)]_0^pi
= (V_m / 2*pi) * [-(-1) - (-1)] = (V_m / 2*pi) * 2 = V_m / pi ~ 0.318 * V_m.
Solution Step 2: Calculate RMS Value V_rms:
(V_rms)^2 = (1 / 2*pi) * int_0^pi V_m^2 * sin^2(theta) d(theta)
Using sin^2(theta) = (1 - cos(2*theta)) / 2:
(V_rms)^2 = (V_m^2 / 4*pi) * int_0^pi (1 - cos(2*theta)) d(theta) = (V_m^2 / 4*pi) * [theta - sin(2*theta)/2]_0^pi
= (V_m^2 / 4*pi) * (pi - 0) = V_m^2 / 4.
Taking square root: V_rms = V_m / 2 = 0.50 * V_m.
Solution Step 3: Compute Form Factor and Peak Factor:
Form Factor = V_rms / V_avg = (V_m / 2) / (V_m / pi) = pi / 2 ~ 1.57.
Peak Factor = V_m / V_rms = V_m / (V_m / 2) = 2.0.

>> 2.5 University Review & Examination Questions

* Q1: Derive the expression for the resonant frequency and Quality factor of an RLC series AC circuit.
Answer: In a series RLC circuit, total impedance is Z = R + j*(omega*L - 1/(omega*C)). Resonance occurs when the net reactive component vanishes: omega_0*L - 1/(omega_0*C) = 0 => omega_0^2 = 1/(LC) => omega_0 = 1/sqrt(LC). Resonant frequency in Hertz: f_0 = 1 / [2*pi*sqrt(LC)]. At this frequency, Z = R, which is minimal, and current I_0 = V/R is maximal. Quality Factor Q is defined as 2*pi * (Maximum energy stored / Energy dissipated per cycle). This equals the ratio of reactive voltage across inductor to applied supply voltage: Q = V_L / V = (I_0 * omega_0 * L) / (I_0 * R) = (omega_0 * L) / R. Substituting omega_0 = 1/sqrt(LC) yields: Q = (1 / R) * sqrt(L / C).
* Q2: Explain the significance of Power Factor in industrial power distribution and detail how shunt capacitors improve it.
Answer: Power factor PF = cos(phi) = P / S. Apparent power S = P / PF. For a fixed active power P, a low lagging power factor forces the electrical supply to deliver vastly higher current (I = S / V). High line current causes: (1) Excessive I^2*R copper

losses in transmission cables and transformers, (2) Large voltage drops (poor regulation), (3) Need for oversized switchgear and thicker cables. Shunt capacitors draw leading reactive current I_C = V * omega * C that cancels the lagging reactive current drawn by inductive motor coils. This reduces total reactive power from Q1 to Q2 without altering active mechanical power P, pulling the power factor close to unity (0.95-1.0) and minimizing total line current.

>> 2.6 Key Takeaways and Revision Points

* RMS represents DC heating equivalence: V_rms = V_m / sqrt(2) ~ 0.707 V_m for pure sine wave.
* Series RLC impedance Z = sqrt(R^2 + (X_L - X_C)^2); inductive reactance leads, capacitive lags.
* Resonant frequency f_0 = 1 / [2*pi*sqrt(LC)]; Quality factor Q = (1/R)*sqrt(L/C) magnifies voltage across reactive elements.
* Power factor PF = cos(phi) = P / S; shunt capacitors supply reactive power Q_C to correct lagging power factor.